

Juan David Muñoz Bolaños

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jmunozbolanos@gmail.com

May 25, 2026

IMS Nanofabrication GmbH

Recruiting Team

Dresdner Str. 47, 1200 Vienna, Austria

Application for the position: Development Engineer

Dear Recruiting Team,

IMS Nanofabrication drives the limits of lithography at the nanometer scale. As an optical systems engineer finishing my PhD at the Medical University of Innsbruck, I have spent the last four years building complex optomechatronic systems (advanced microscopes) and developing data analysis pipelines to optimize their performance. My background in physics engineering, combined with my exposure to industrial semiconductor workflows at the ASML and ZEISS summer schools, makes me eager to contribute to the development of your Multi Beam Mask Writer systems.

In my current research, I designed and built an automated wavefront sensing and adaptive optics system from scratch. This involved integrating high precision optical components, spatial light modulators, and scientific cameras, while writing the entire control and data acquisition stack in Python and C++. To ensure the system achieved the best potential performance, I developed iterative phase retrieval and optimization algorithms to process wavefront data in real time. This hands-on experience in combining hardware control with advanced signal processing directly aligns with your need for validating new features and analyzing complex log and metrology data.

Beyond system development, I focus on structured testing and quality frameworks. I have structured my instrument development following V-Model systems engineering principles, ensuring clear requirements tracking from initial concept to final validation. By applying Design of Experiments and systematic optical tolerance analysis in python, I have successfully bridged the gap between early stage R&D and stable, reproducible system performance.

I will complete my PhD in June 2026. I am Colombian originally and hold an unrestricted working resident permit in Austria (rot-weiss-rot-plus). I look forward to discussing how my background in optomechanical and electronic systems, data analysis, and structured testing can support your team in Vienna.

Best regards,

Juan David Munoz Bolanos

Juan David Munoz Bolanos

Junior Optical System Engineer

Rechenauerstraße 42, Innsbruck, Austria

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Summary

Junior system engineer with hands-on lab work building, aligning, and calibrating optical laser systems using spatial light modulators and holography. I'm comfortable with PC and control technology, used to digging into technical problems systematically.

Research/Professional Experience

Jan 2022 to Jun 2026 **Medical University of Innsbruck** *Innsbruck, Austria*
Optical Systems Engineer / PhD Researcher

Optomechatronic System Design: Designed and built a multiphoton microscope and an automated wavefront sensing system from scratch, integrating lasers, spatial light modulators, and deformable mirrors.

Data Analysis & Algorithms: Developed Python and C++ pipelines for real time signal processing, iterative phase retrieval, and wavefield simulations to optimize system performance.

Systems Engineering & Quality: Applied V Model methodologies, Design of Experiments, and FMEA to track requirements and systematically resolve technical defects during prototype iterations.

Lab Management & Safety: Managed a high power laser laboratory, ensuring EH&S compliance, maintaining equipment logs, and coordinating component sourcing with suppliers like Thorlabs and Edmund Optics.

Aug 2018 to Oct 2019 **INTECOL S.A.S.** *Medellín, Colombia*
Junior Machine Vision Engineer (Field Service)

Machine Vision Pipelines: Engineered and deployed Halcon and Cognex VisionPro image processing pipelines for high speed industrial production lines, improving quality control.

Illumination Design: Designed custom bright field and dark field illumination systems to reliably detect defects, liquid levels, and sealing states on fast moving products.

Industrial Automation: Integrated vision systems with robotic arms and PLC controllers, executing on site testing, validation, and customer sign off.

Technical Competencies

Software & Tools

Python C++ LabVIEW Halcon OpenCV ZEMAX MATLAB Git

Optical Systems

Adaptive Optics Wavefront Sensing Microscopy Spectroscopy Laser Systems
DMDs

Systems & Quality

V-Model 8D Problem Solving Design of Experiments ISO 9001

Education

Jan 2022 to Jun 2026 **Medical University of Innsbruck** *Innsbruck, Austria*
Doctor of Philosophy in Optical Metrology

Oct 2019 to Sep
2021

Friedrich Schiller Univ. Jena
Master of Science in Photonics

Jena, Germany

Aug 2012 to Aug
2018

University of Cauca
Bachelor of Science in Physics Engineering

Popayán, Colombia

Publications

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Adaptability & Independent Working: Self-driven approach to bringing concepts to working systems; comfortable in dynamic, fast-paced environments.

Teamwork & Communication: Cross-functional collaboration across optics, hardware, software, and production teams; structured root cause analysis.

Personal Interests

Biking/Hiking

Bachata Dancing

Harmonica

Learning German

Languages

English (Fluent) · German (Intermediate B2) · Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte
Medical University of Innsbruck
monika.ritsch-marte@i-med.ac.at

PD Dr. Christoph Krafft
Leibniz IPHT
christoph.krafft@leibniz-ipht.de

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

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REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garces
TRADUCTOR e INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

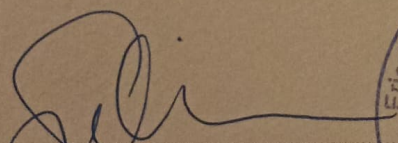
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

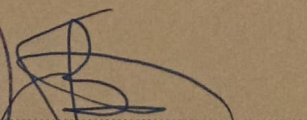
Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

ASML

PhD Master Class

Dear participant of the ASML PhD Master Class | Spring edition 2026,

We're looking forward to meeting you on March 19, 20 & 21.

Here's an information package containing details about the masterclass program. Please read this carefully so you know what to expect and how to prepare for it.

To get the most out of this masterclass for you, we want your input on what you would like to learn. At the end of this briefing, there is a link where we ask you to share your preferences with us.

Do not hesitate to contact us if you still have questions after this briefing.

Laura de Swart
Program Manager, ASML
+31 6 1562 7555

You will be joining us live at ASML HQ in Veldhoven on Friday and Saturday. Make sure you bring your **valid ID** to enter the ASML buildings. Without it, you will **not** be allowed inside. On Thursday we will meet at the Philips Stadium in the center of Eindhoven for an informal start to the event. The dress code for the entire masterclass is informal.

Program

Below is the program for the three days again. On the next pages, you will find more details on the different parts of the program.

PhD Master Class

Thursday, March 19

19:30 – 22:30 Informal activity @ Philips Stadium

Friday, March 20

08:30 – 09:00	Welcome
09:00 – 10:30	Introduction into ASML Technology
10:30 – 10:50	Break
10:50 – 12:00	Pitches from former participants
12:00 – 13:00	Lunch break
13:00 – 18:00	Working on cases
18:00 – 21:45	Dinner & presentations & feedback on case results
21:45 – 22:00	Closing day one

Saturday, March 21

08:30 – 09:00	Welcome back
09:00 – 12:50	Carousel • Matching conversations • HR corner • Q&A • ASML tours
12:50 – 13:50	Lunch break
13:50 – 14:00	Wrap up and closing

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stammer

Dr. Thomas Stammer
CTO & Head of Product Strategy



